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CHAPTER 5: ADVANCED ENCRYPTION STANDARD

TRUE OR FALSE

T F 1. AES uses a Feistel structure.

T F 2. At each horizontal point, **State** is the same for both encryption and decryption.

T F 3. DES is a block cipher intended to replace AES for commercial applications.

T F 4. The nonlinearity of the S-box is due to the use of the multiplicative inverse.

T F 5. Virtually all encryption algorithms, both conventional and public-key, involve arithmetic operations on integers.

T F 6. Compared to public-key ciphers such as RSA, the structure of AES and most symmetric ciphers is quite complex and cannot be explained as easily as many other cryptographic algorithms.

T F 7. InvSubBytes is the inverse of ShiftRows.

T F 8. The ordering of bytes within a matrix is by column.

T F 9. In the Advanced Encryption Standard the decryption algorithm is identical to the encryption algorithm.

T F 10. The S-box is designed to be resistant to known cryptanalytic attacks.

T F 11. As with any block cipher, AES can be used to construct a message authentication code, and for this, only decryption is used.

T F 12. The inverse add round key transformation is identical to the forward add round key transformation because the XOR operation is its own inverse.

T F 13. The Rijndael developers designed the expansion key algorithm to be resistant to known cryptanalytic attacks.

T <u>F</u> 14. The transformations AddRoundKey and InvMixColumn alter the sequence of bytes in State.

<u>T</u> F 15. AES can be implemented very efficiently on an 8-bit processor.

MULTIPLE CHOICE

1. The Advanced Encryption Standard was published by the ________ in 2001.

A. ARK B. FIPS

C. IEEE D. NIST

2. In Advanced Encryption Standard all operations are performed on ________ bytes.

A. 8-bit B. 16-bit

C. 32-bit D. 4-bit

3. The AES cipher begins and ends with a(n) ________ stage because any other stage, applied at the beginning or end, is reversible without knowledge of the key and would add no security.

A. Substitute bytes B. AddRoundKey

C. MixColumns D. ShiftRows

4. A ________ is a set in which you can do addition, subtraction, multiplication and division without leaving the set.

A. record B. standard

C. field D. block

5. Division requires that each nonzero element have a(n) ________ inverse.

A. multiplicative B. divisional

C. subtraction D. addition

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6. In AES, the arithmetic operations of addition, multiplication and division are performed over the finite field _______ .

   A. Z_p B. $a/b = a(b^{-1})$

   C. GF(2^{n-1}) D. $\boxed{\text{GF}(2^8)}$

7. In the AES structure both encryption and decryption ciphers begin with a(n) ________ stage, followed by nine rounds that each include all four stages, followed by a tenth round of three stages.

   A. Substitute bytes B. $\boxed{\text{AddRoundKey}}$

   C. MixColumns D. ShiftRows

8. The final round of both encryption and decryption of the AES structure consists of ________ stages.

   A. one B. two

   C. four D. $\boxed{\text{three}}$

9. The first row of State is not altered; for the second row a 1-byte circular left shift is performed; for the third row a 2-byte circular left shift is performed; and for the fourth row a 3-byte circular left shift is performed. This transformation is called ________ .

   A. AddRoundKey B. $\boxed{\text{ShiftRows}}$

   C. MixColumns D. Substitute bytes

10. In the AddRoundKey transformation the 128 bits of State are bitwise XORed with the _______ of the round key.

   A. 256 bits B. $\boxed{\text{128 bits}}$

   C. 64 bits D. 512 bits

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11. The ________ is when a small change in plaintext or key produces a large change in the ciphertext.

A. avalanche effect
B. Rcon
C. key expansion
D. auxiliary exchange

12. The encryption round has the structure:

A. ShiftRows, MixColumns, SubBytes, InvMixColumns
B. SubBytes, ShiftRows, MixColumns, AddRoundKey
C. MixColumns, ShiftRows, SubBytes, AddRoundKey
D. InvShiftRows, InvSubBytes, AddRoundKey, InvMixColumns

13. ________ affects the contents of bytes in State but does not alter byte sequence and does not depend on byte sequence to perform its transformation.

A. InvSubBytes
B. ShiftRows
C. SubBytes
D. InvShiftRows

14. In the general structure of the AES encryption process the input to the encryption and decryption algorithms is a single ________ block.

A. 32-bit
B. 256-bit
C. 128-bit
D. 64-bit

15. The cipher consists of *N* rounds, where the number of rounds depends on the ________ .

A. key length
B. output matrix
C. State
D. number of columns

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SHORT ANSWER

1. The AES is a block cipher intended to replace DES for commercial applications. It uses a 128-bit block size and a key size of 128, 192, or 256 bits.

2. The four separate functions of the Advanced Encryption Standard are: permutation, arithmetic operations over a finite field, XOR with a key, and byte substitution.

3. The National Institute of Standards and Technology chose the _________ design as the winning candidate for AES.

4. The cipher consists of N rounds, where the number of rounds depends on the key length.

5. AES processes the entire data block as a single matrix during each round using substitutions and permutation.

6. The first N - 1 rounds consist of four distinct transformation functions: SubBytes, ShiftRows, AddRoundKey, and MixColmns .

7. The forward substitute byte transformation, called SubByte, is a simple table lookup.

8. The MixColmns transformation operates on each column individually. Each byte of a column is mapped into a new value that is a function of all four bytes in that column.

9. The mix column transformation combined with the AddRoundKey transformation ensures that after a few rounds all output bits depend on all input bits.

10. The AES key expansion algorithm takes as input a four-word (16-byte) key and produces a linear array of 44 words (176 bytes).

11. The standard decryption round has the structure InvShiftRows, InvSubBytes, AddRoundKey , InvMixColumns.

12. InvShiftRaws affects the sequence of bytes in State but does not alter byte contents and does not depend on byte contents to perform its transformation.

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13. A more efficient implementation can be achieved for a 32-bit processor if operations are defined on 32-bit words.

14. An example of a finite field is the set Z_p consisting of all the integers {0, 1, . . . , *p* - 1}, where *p* is a ________ and in which arithmetic is carried out modulo *p*.

15. A polynomial *m(x)* is called ________ if and only if *m(x)* cannot be expressed as a product of two polynomials, both of degree lower than that of *m(x)*.